

Multiple Sclerosis Prediction Based on Clinical/Biological Data

Introduction

Multiple sclerosis is a chronic, autoimmune disease in which the central nervous system (CNS) is impacted. For patients with MS, the immune system mistakenly targets the myelin sheath around nerve cells. As a result, inflammation and scarring—aka *sclerosis*—occurs at regions called *lesions*. Researchers use the Expanded Disability Status Scale (EDSS) to evaluate the level of disability in patients, with 0 being no disability and 10 being death due to MS (Trust). The scale is based on eight neurological functions, including vision, coordination, strength, sensation, bladder function, and cognition, as well as the ability to walk (Trust). Currently, EDSS of MS patients remains robust and unpredictable. It is imperative to continue researching the disease, especially as the quality-of-life of patients with MS significantly decreases (Koftuniuk et al.).

Although there is no cure to the disease, various medicines can be taken to alleviate some of its clinical symptoms. However, recent research suggests that these medicines may have varying impacts on the quality-of-life of MS patients (Hirt et al.). Computational work regarding MS has confirmed that indeed, deep learning to predict EDSS can reach accuracies of up to 91.2%. Nevertheless, such models are trained on segmented lesions from MRI imaging data (Alves et al.). Not only are consensus on segmentations difficult to obtain (as they require multiple consultations from neurologists), but MRIs can also be costly or inaccessible.

In this paper, the question at hand is: how well can clinical data versus biological marker data predict EDSS, and, more broadly, MS? The reason I focus on this question is because

clinical data, like age or gender have been shown to be potential confounding variables in MS diagnosis. For example, women are much more likely to be diagnosed with MS, and typically present more severe symptoms than men (Mowry). As such, I hypothesize that they will be effective at predicting EDSS/MS. Then, I looked at potential causality using biological marker data. Though these markers have not officially been declared as part of the pathology of MS—given that it is still an emerging field of research—these markers are occasionally used in the laboratory-based diagnosis of MS.

Overall, I use both random forest regression (for predicting EDSS, which is a numerical variable) and classification (for predicting MS, which is a binary present/not present variable). The random forest models were trained on three subsets of data: 1) clinical data, 2) biological data, and 3) data with mixed effects—i.e. a combination of both clinical and biological data. The goal of this structure is to determine which data is most relevant for EDSS/MS prediction, and ultimately provide consideration for the type of data that future MS models will be trained on.

Methodology and Model Selection

The first task—EDSS prediction—is a regression task, since we’re predicting EDSS on a scale of 0 to 10. EDSS is typically nonlinear, as progression typically intensifies with age. Nonlinear regression, however, is limited because there will be values that predict EDSS values off the scale (such as negative or above 10). Therefore, I chose to use random forests. This is because they capture high-dimensional data and nonlinear relationships well (as opposed to, say, Bayesian logistic regression). Our dataset is limited because there are only 60 patients, making it unsuitable for neural network training. Also, because the dataset had 20 variables, there were some entries with unknown values, and random forests are more robust to this noise. Lastly,

tree-based algorithms are advantageous because they efficiently handle mixed data types (*What Is Random Forest?*).

On the other hand, our second task—MS prediction—is a classification task. We classify data points as either non-MS or MS presenting patients. Since MS prediction will be trained on the same type of data as EDSS prediction, I chose to continue using a random forest model.

Further, if the tree-based algorithm performs well on EDSS regression, it is likely to also perform well on MS classification.

Data Description

The first dataset is collected from Muslim et. al's paper titled, "Brain MRI dataset of multiple sclerosis with consensus manual lesion segmentation and patient meta information." Patients included in this dataset are confirmed to have multiple sclerosis prior to metadata information collection. It is of note that patients were solely being treated at the Baghdad Teaching Hospital, located in Baghdad, Iraq. In total, there are 60 patients. The original data's variables are: gender, age, age of onset, EDSS, time difference between MRI acquisition and EDSS determination, types of medicines, presenting symptoms, comorbidity, and various qualities for abnormal functions. These functions include: pyramidal, cerebella, brain stem, sensory, sphincters, visual, mental, speech, motor system, sensory system, coordination, gait, bowel and bladder function, mobility, mental state, optic discs, fields, nystagmus, ocular movement, and swallowing.

Since data for some variables was limited and biased—meaning its possible values (present versus not present) were unbalanced—the data being used for our model focuses on a narrower spectrum of variables. Namely, we focus on: gender, age, age of onset, types of

medicines, presenting symptoms, and comorbidity to predict EDSS. Age of onset refers to the age in which a patient first develops symptoms of multiple sclerosis. Comorbidity refers to the simultaneous presence of two or more diseases. While some variables were numerical—like age, age of onset, and EDSS—other variables were categorical. We used one-hot encoding to handle these categorical values for nonlinear regression. First, types of medicine included Golenya, Tysabri, Avonex, Betaferon, and Rebif. These variables were encoded to 0, 1, 2, 3, and 4, respectively. Second, gender was encoded to 0 for male and 1 for female. Third, lack of comorbidity was encoded to 0, while presence was encoded to 1. Fourth, the types of symptoms presented were motor, sensory, visual, balance, or comorbidity (patients could accumulate a combination of multiple, or just have a singular symptom). To preprocess the symptoms, I split the different types into columns. Then, I encoded them. The presence of a symptom is indicated with a 1, and the absence of a symptom is indicated with a 0. Overall, the Muslim et. al dataset will be u

Gender	Age	Age of onset	Types of Medicines	Motor	Sensory	Visual	Balance	Comorbidity	EDSS
1	56	43	0	1	0	0	0	0	3
1	29	19	0	0	1	0	0	0	1.5
1	15	8	1	1	0	0	0	0	4

Example of rows in Muslim et. al dataset

Additionally, I wanted my analysis to be more robust, and account for any confounding variables (such as the diagnosis/measurement of patients with EDSS potentially being skewed for that specific hospital). For this reason, I looked at a second dataset collected from Kaggle. This dataset was collected from the National Institute of Neurology and Neurosurgery (NINN) in Mexico City, and included both patients with and without diagnosis. Unlike the first dataset,

which was purely clinical patient metadata, the second dataset also includes biological marker data.

Variable	Type	Possible Values	Skew
Gender	Categorical	1=male, 2=female	No
Age	Numerical	20 to 80	Yes
Schooling	Numerical	0 to 25	No
Breastfeeding	Categorical	1=yes, 2=no, 3=unknown	No
Varicella	Categorical	1=positive, 2=negative, 3=unknown	No
Initial symptoms	Categorical	1=visual, 2=sensory, 3=motor, 4=other, 5=visual and sensory, 6=visual and motor, 7=visual and others, 8=sensory and motor, 9=sensory and other, 10=motor and other, 11=Visual, sensory and motor, 12=visual, sensory and other, 13=Visual, motor and other, 14=Sensory, motor and other, 15=visual,sensory,motor and other	No
Mono or polysymptomatic	Categorical	1=monosymptomatic, 2=polysymptomatic, 3=unknown	Yes
Oligoclonal bands	Categorical	0=negative, 1=positive, 2=unknown	Yes
Lower limb somatosensory evoked potentials	Categorical	0=negative, 1=positive	Yes

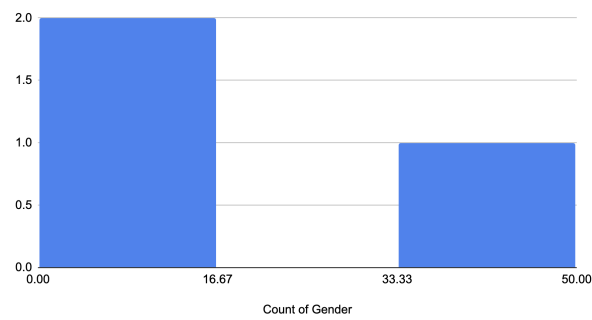
(LLSEP)			
Upper limb somatosensory evoked potentials (ULSEP)	Categorical	0=negative, 1=positive	Yes
Visual evoked potentials (VEP)	Categorical	0=negative, 1=positive	Yes
Brainstem auditory evoked potentials (BAEP)	Categorical	0=negative, 1=positive	Yes
Periventricular MRI	Categorical	0=negative, 1=positive	No
Cortical MRI	Categorical	0=negative, 1=positive	Yes
Infratentorial MRI	Categorical	0=negative, 1=positive	Yes
Spinal cord MRI	Categorical	0=negative, 1=positive	Yes
Initial EDSS	Numerical	0 to 3	Yes
Group	Categorical	1=MS, 2=non-MS	Yes

Variables from Kaggle dataset

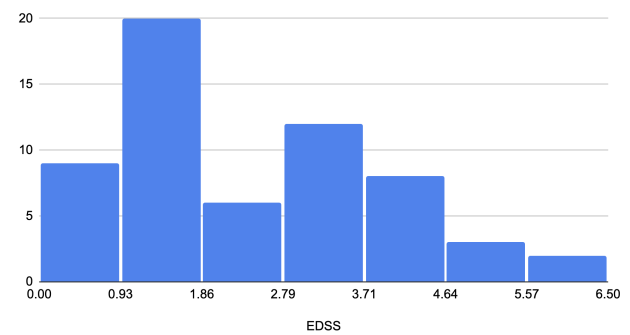
Data Analysis

For the Muslim et. al dataset, there was little to no skewing. However, there were two notable skewed variables: gender and EDSS. There were typically more female data points than male, which aligns with previous conclusions about females being more impacted by MS. EDSS was additionally positively skewed; this may be justified by the lower prevalence of more extreme cases. Furthermore, age and age of onset had no predictable patterns.

Histogram of Count of Gender

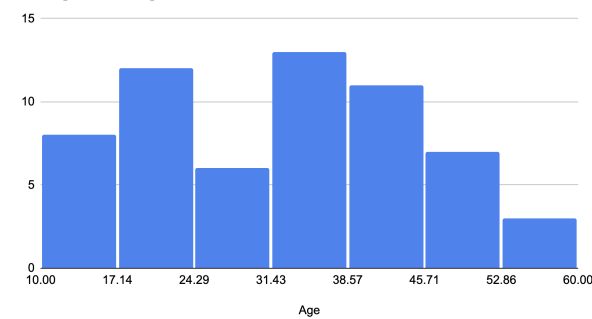


Histogram of EDSS

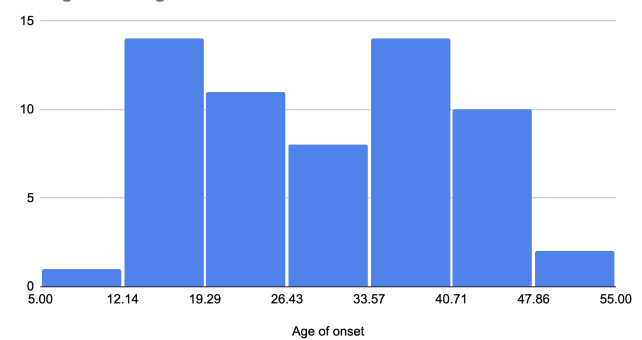


Skewed variables in Muslim et. al dataset

Histogram of Age

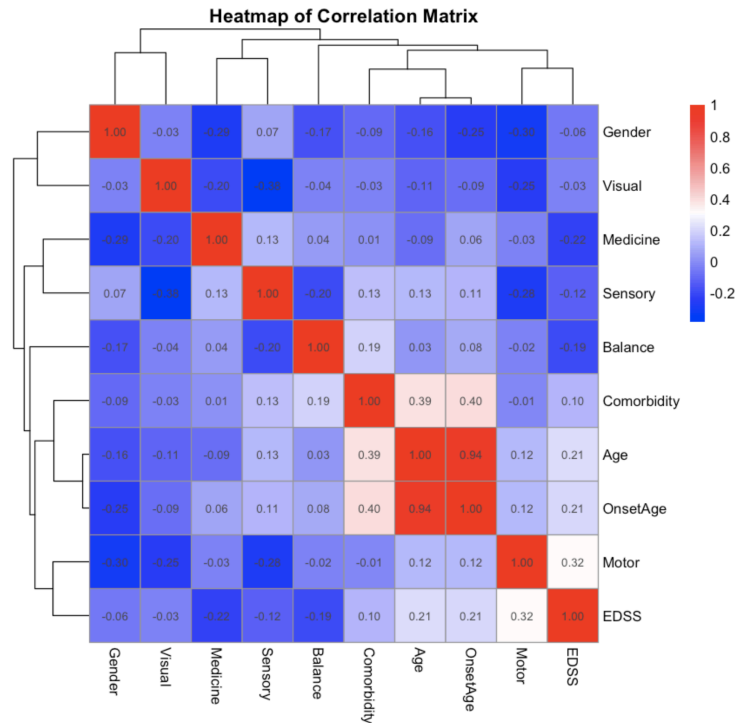


Histogram of Age of onset



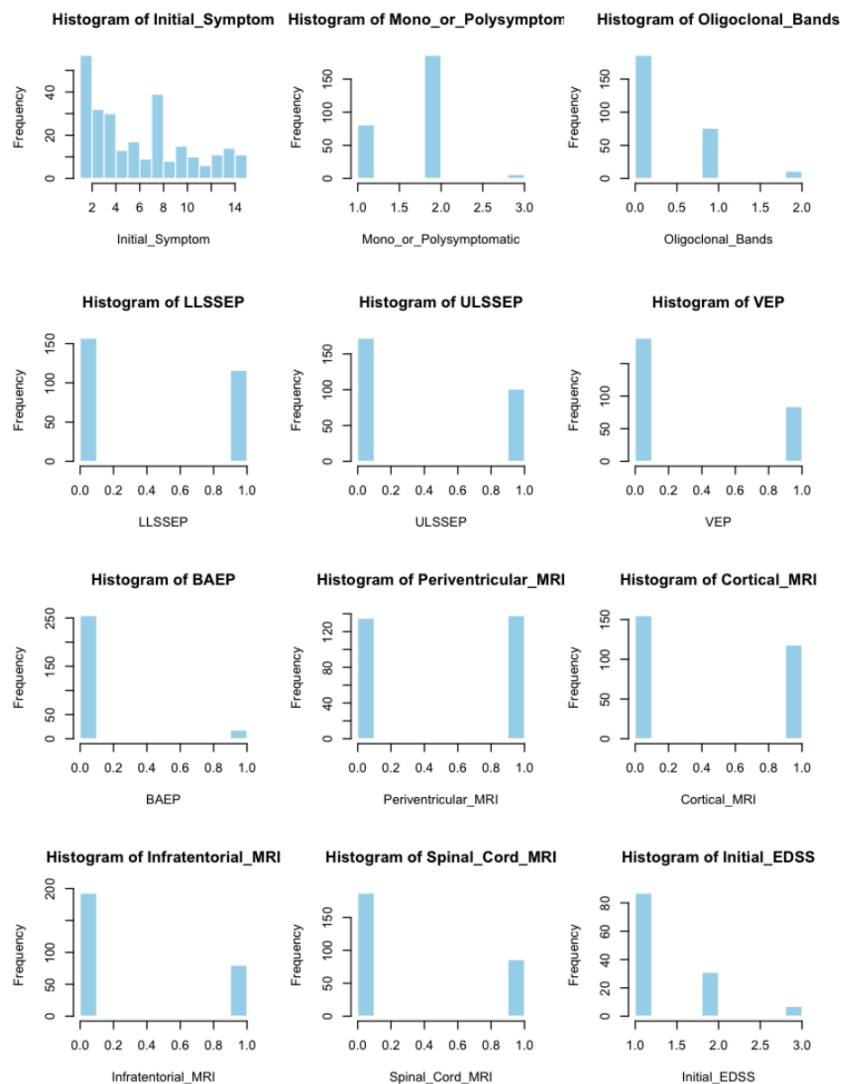
Randomly distributed age variables in Muslim et. al dataset

Moreover, the randomness of the Muslim et. al dataset is exemplified in its correlation matrix. The correlation matrix displayed that there were nearly no variables that were highly correlated to each other. The exception to this trend is the age of onset and age of the patient. This result makes practical sense—older patients can be diagnosed later, and younger patients tend to be diagnosed at an earlier age.



Correlation matrix for variables in Muslim et. al dataset

For the Kaggle dataset, there were not many clinically skewed variables with the exception of age and mono/poly-symptomatic. It is most interesting that nearly all biological data was negatively skewed despite an even split in non-MS/MS patients. Such a result may be justified due to the holistic diagnosis of MS; patients do not need to present all clinical or biological symptoms in order to be diagnosed. Thus, despite an even split in non-MS/MS, a subset of MS patients may still not present a biological symptom. However, indications of MS from periventricular MRI were *not* skewed, and was a nearly even split. This may indicate that, despite other biological symptoms not necessarily being prevalent in the entire MS population, an abnormal periventricular MRI is a *hallmark* in MS patients.

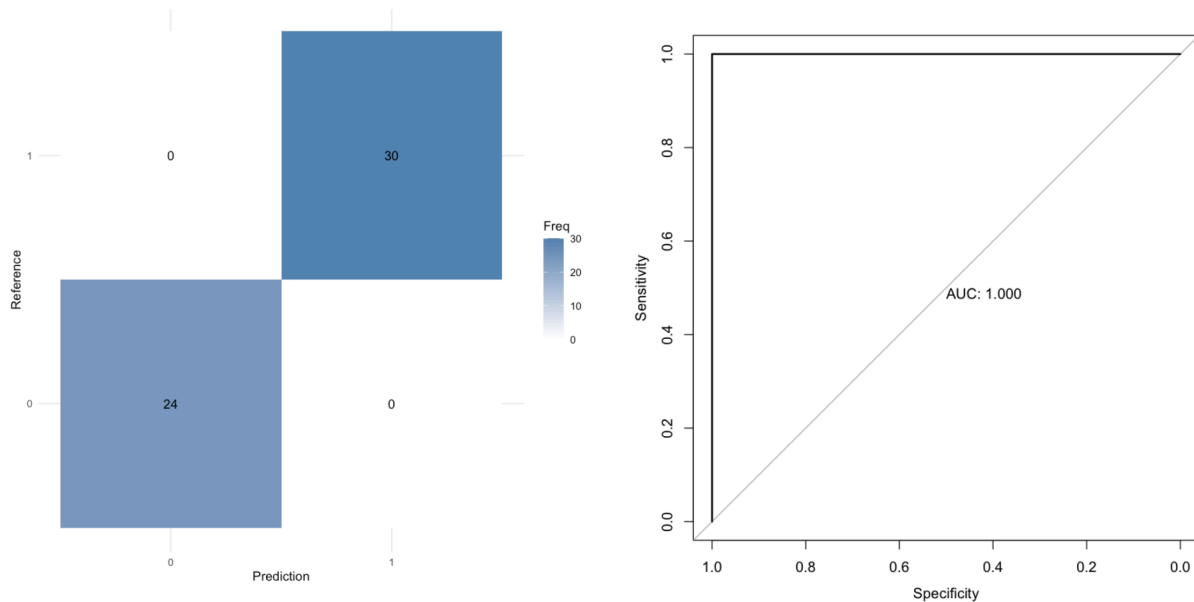


Subset of data distributions in Kaggle dataset—note that the number of symptoms is skewed, and that most biological data (with the exception of periventricular MRI data) is negatively skewed indicating a lack of presence

Results

For MS prediction (which was only performed on the Kaggle dataset, which had both non-MS/MS patients), a random forest model achieved an accuracy of 100% and AUC of 1. Looking at the confusion matrix of the model, there are no false positives or negatives. A high

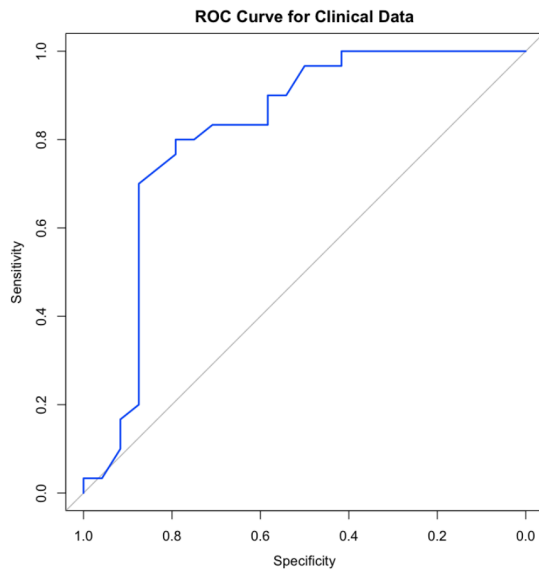
accuracy may indicate that the model is overfitting. Nevertheless, it is important to note that the dataset consisted of 54 data points after removing data points with any null values. Therefore, a perfect accuracy on a smaller dataset may correlate to a high-performing model on a larger dataset.



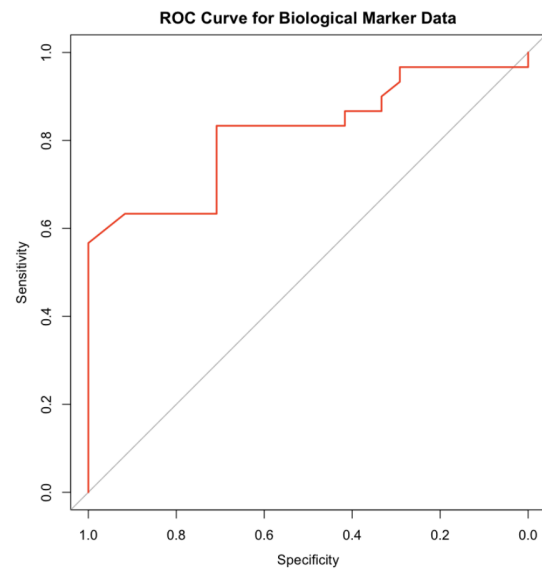
Confusion matrix and AUC curve of random forest model for MS classification on full Kaggle dataset

Next, I divided the full Kaggle dataset into two subsets: a clinical subset and a biological subset. Since the random forest model performed well on the mixed-effect data, I hypothesized that it would also perform well on the isolated data. Surprisingly, this was not the case—instead, accuracy dramatically decreased. The model trained on clinical data achieved an accuracy of 79%, whereas the model trained on biological data achieved an accuracy of 70%. Though, looking at the AUC curve, both models learn at similar rates and ultimately achieve a similar AUC of roughly 0.81.

AUC for Clinical Data: 0.8180556



AUC for Biological Marker Data: 0.8173611



AUC curves of random forest model for MS classification on isolated Kaggle dataset: clinical and biological marker data, respectively

Next, for EDSS regression, this task proved to be more unpredictable. Since random forests performed well for MS prediction, and the data was in the same structure, I attempted to use random forests for EDSS regression. On the Kaggle dataset, with mixed effect variables, a random forest achieved an R-squared value of 0.9653313 and MSE of 0.02508908. Below are the predictors in terms of increasing node purity, where the higher the number the greater impact the predictor had in making the decision. Significant predictors included: age, schooling, and infratentorial MRI. These results are especially intriguing given that age, as shown in the Muslim et. al dataset, was normally distributed in MS patients. Schooling being a significant predictor can likely be explained by its high correlation with age, where older patients tend to have more years of schooling. Lastly, infratentorial MRI being a significant predictor contradicts the initial data analysis. Initially, abnormal periventricular MRI data was implied to be a hallmark of MS patients.

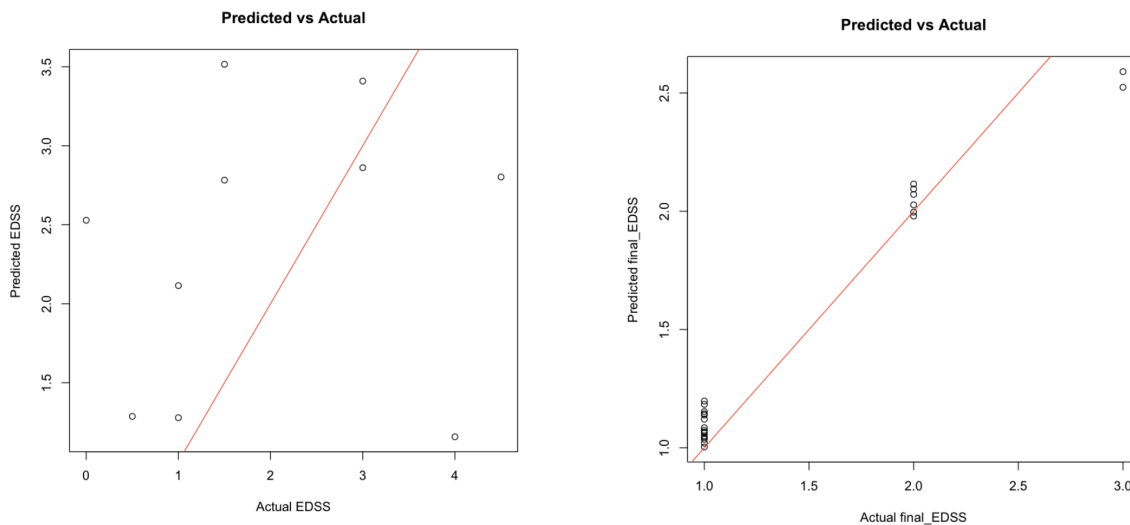
Gender	0.1907603
Age	2.2087756
Schooling	1.5747309
Breastfeeding	0.5134601
Varicella	0.2554599
Initial_Symptom	1.0919332
Mono_or_Polysymptomatic	0.1834432
Oligoclonal_Bands	0.1843359
LLSSEP	0.3734697
ULSSEP	0.5219844
VEP	0.2750041
BAEP	0.1195429
Periventricular_MRI	0.5918045
Cortical_MRI	0.1549707
Infratentorial_MRI	1.5877784
Spinal_Cord_MRI	0.4409631

Significant predictors, as determined by random forest model for EDSS regression trained on Kaggle dataset

Again, the trend with the mixed effect dataset performing better was also true for the EDSS regression. When trained on purely clinical data from the second dataset, I achieved an R-squared value of 0.04713224. Trained on the biological marker data, I achieved an R-squared value of 0.05. To make sure that this wasn't just an artifact of the second dataset, I returned to the first dataset that was purely clinical. Indeed, the clinical data performed very poorly on the first dataset, with an R-squared value of 0.02.

	Kaggle Dataset	Muslim et. al Dataset		
R-squared	Clinical	Clinical and biological	Clinical	Biological
	0.02	0.965	0.047	0.05

Varying performances of random forest model for EDSS regression trained on datasets/data subsets



Scatterplot of actual EDSS versus predicted EDSS based on clinical data, showing low accuracy (left); based on clinical and biological data, showing high accuracy (right)

Conclusions

As shown through the different datasets and approaches I used in trying to predict EDSS, it is very difficult to analyze and has nearly no pattern with the clinical and biological variables taken separately—at least the ones used in this project. There may be some clinical implications for this, since there have been recent debates about whether EDSS is an appropriate scale. EDSS is measured based on eight neurological functions, including vision, coordination, strength, sensation, bladder function, cognition, and the ability to walk. Capturing the disability scale with these eight functions can obviously be very limiting. Besides the limitations of the simplified functional attributes to quality of life, the scoring of EDSS has also been highly unreliable. For example, inter-rater variability in EDSS was as high as 30% for the same patient (Cohen et al.). However, when we take the mixed effect dataset, random forests perform strongly for both diagnosing MS binary classification and EDSS regression.

Previously, I briefly touched upon periventricular MRI versus infratentorial MRI data. These models suggest that infratentorial lesions are more informative for predicting conversion, even though periventricular lesions are more prevalent. Overall, using a mixed effect dataset to train random forests consistently performed strongly—and outperformed separate data subsets—for both diagnosing MS binary classification and EDSS regression. Future studies should focus on such a holistic standpoint for prediction, rather than isolating clinical or biological marker data; the complex interactions between these heterogeneous variables are strongly predictive of MS/EDSS.

Bibliography

- Alves, Pedro, et al. “Validation of a Machine Learning Approach to Estimate Expanded Disability Status Scale Scores for Multiple Sclerosis.” *Multiple Sclerosis Journal - Experimental, Translational and Clinical*, vol. 8, no. 2, June 2022, p. 20552173221108635. *PubMed Central*, <https://doi.org/10.1177/20552173221108635>.
- Cohen, Mikael, et al. “Should We Still Only Rely on EDSS to Evaluate Disability in Multiple Sclerosis Patients? A Study of Inter and Intra Rater Reliability.” *Multiple Sclerosis and Related Disorders*, vol. 54, Sept. 2021, p. 103144. *ScienceDirect*, <https://doi.org/10.1016/j.msard.2021.103144>.
- Harati Kabir, Vida, et al. “Prediction of Expanded Disability Status Scale in Patients with MS Using Deep Learning.” *Computers in Biology and Medicine*, vol. 182, Nov. 2024, p. 109143. *ScienceDirect*, <https://doi.org/10.1016/j.compbiomed.2024.109143>.
- Hirt, Julian, et al. “Clinical Trial Evidence of Quality-of-Life Effects of Disease-Modifying Therapies for Multiple Sclerosis: A Systematic Analysis.” *Journal of Neurology*, vol.

271, no. 6, June 2024, pp. 3131–41. *PubMed*,

<https://doi.org/10.1007/s00415-024-12366-5>.

Kołtuniuk, Aleksandra, et al. “The Quality of Life in Patients with Multiple Sclerosis – Association with Depressive Symptoms and Physical Disability: A Prospective and Observational Study.” *Frontiers in Psychology*, vol. 13, Jan. 2023, p. 1068421. *PubMed Central*, <https://doi.org/10.3389/fpsyg.2022.1068421>.

Mowry, Ellen. “Multiple Sclerosis: Why Are Women More at Risk?” *Johns Hopkins Medicine*, 9 Feb. 2023, <https://www.hopkinsmedicine.org/health/conditions-and-diseases/multiple-sclerosis-ms/multiple-sclerosis-why-are-women-more-at-risk>.

“Multiple Sclerosis Disease.” *Kaggle*, 2023, <https://www.kaggle.com/datasets/desalegngeb/conversion-predictors-of-cis-to-multiple-sclerosis>.

Muslim, Ali M., et al. “Brain MRI Dataset of Multiple Sclerosis with Consensus Manual Lesion Segmentation and Patient Meta Information.” *Data in Brief*, vol. 42, June 2022, p. 108139. *ScienceDirect*, <https://doi.org/10.1016/j.dib.2022.108139>.

Omarzai, Fraidoon. “XGBoost Regression In Depth.” *Medium*, 25 Mar. 2025, <https://medium.com/@fraidoonomarzai99/xgboost-regression-in-depth-cb2b3f623281>.

Trust, M. S. *Expanded Disability Status Scale (EDSS) | MS Trust*. <https://mstrust.org.uk/a-z/expanded-disability-status-scale-edss>. Accessed 27 Apr. 2025.

What Is Random Forest? 20 Oct. 2021, <https://www.ibm.com/think/topics/random-forest>.

XGBoost Documentation — Xgboost 3.0.0 Documentation.

https://xgboost.readthedocs.io/en/release_3.0.0/. Accessed 27 Apr. 2025.